

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 1

Name	Chew Chiu Xian
IC No. / Matric No.	A23CS0061
Year / Program	2 SECPH
Section	02
Lecturer Name	Dr. Seah Choon Sen
Case Study	Saving Management in KADA System
System Name	Koperasi KADA Online Platform
Group Name	Tech EcoPioneer

Table of Contents

SECTION A (SYSTEM'S OVERVIEW)

1.1 Overall Description	3
1.2 Project Weaknesses or Improvement	3
1.3 Proposed Module	3

SECTION B (DATABASE PLANNING AND DESIGN)

2.1 Conceptual ERD	4
2.2 Logical ERD	5
2.3 Data Dictionary for Logical ERD	
2.3.1 Identify Entity	6
2.3.2 Attributes Description	6
2.4 Relational Database Schema	8
2.5 Normalization	
2.5.1 Member Relation	9
2.5.2 Account Relation	11
2.5.2 Saving Account Relation	12
2.5.3 Recurring Payment Relation	13
2.5.4 Savings Goals Relation	16
2.5.5 Transaction Relation	17

SECTION A (SYSTEM'S OVERVIEW)

1.1 Overall Description

Group Members' Name	Matric No	Module
Dheshighan A/L Saravana Moorthy	A23CS0072	Login
Evelyn Goh Yuan Qi	A23CS0222	Member Management
Chew Chiu Xian	A23CS0061	Savings Management
Ng Yu Hin	A23CS0148	Loans Management
Lau Yan Kai	A23CS0098	Reports Management

The **Koperasi KADA Online Platform** is a cloud-based financial system designed to make cooperative operations easier. It automates tasks like member registration, savings management, loan applications, and reporting. The platform offers a simple interface where members can manage their savings, apply for loans, and set up regular payments. Administrators can approve applications, validate documents, and generate reports. The system also includes helpful features like OTP login, automatic notifications, and a chatbot for assistance.

The **Savings Module**, which I'm working on, allows members to create savings goals, set up regular payments, and track their progress. Its goal is to make managing finances easier for members and reduce the workload for staff.

1.2 Project Weakness or Improvement

The existing KADA system suffers from several issues, including:

- **Lack of Integration:** Manual processes for savings and transactions.
- **Security Concerns:** Limited security features, making the system vulnerable to unauthorized access.
- **Lack of Accessibility:** No online access, making it inconvenient for members in remote areas.
- **Static Member Engagement:** Members cannot track their savings or set goals digitally.

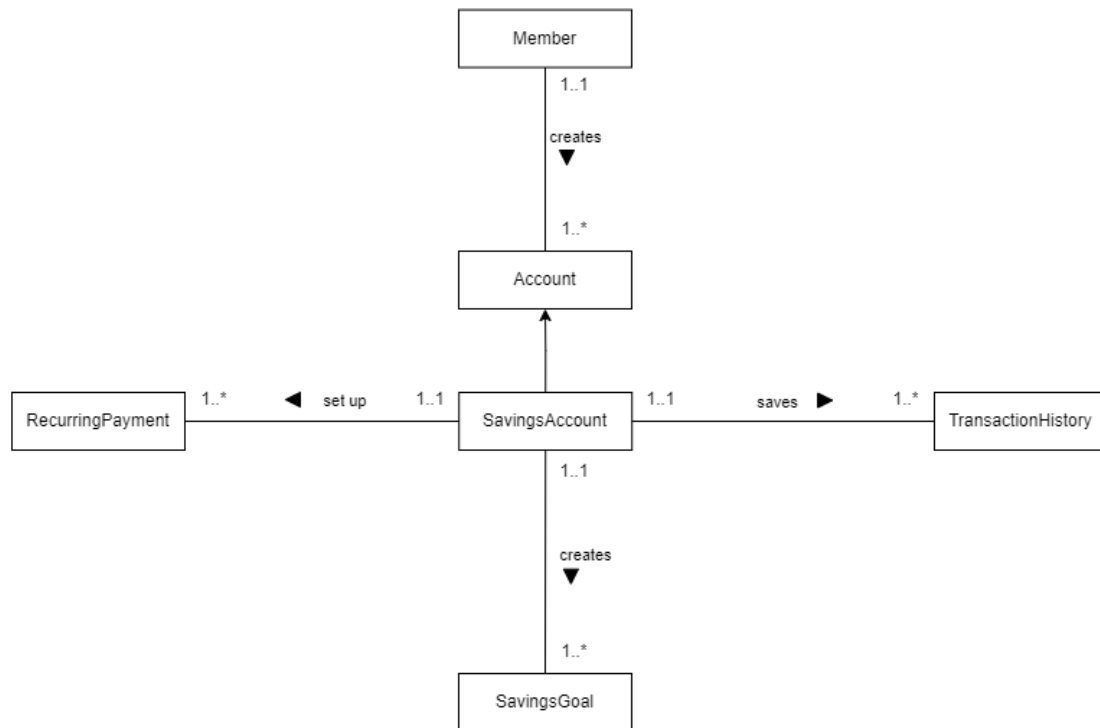
1.3 Proposed Module

To address these weaknesses, I propose a new module called "**Savings Management**". This module will include:

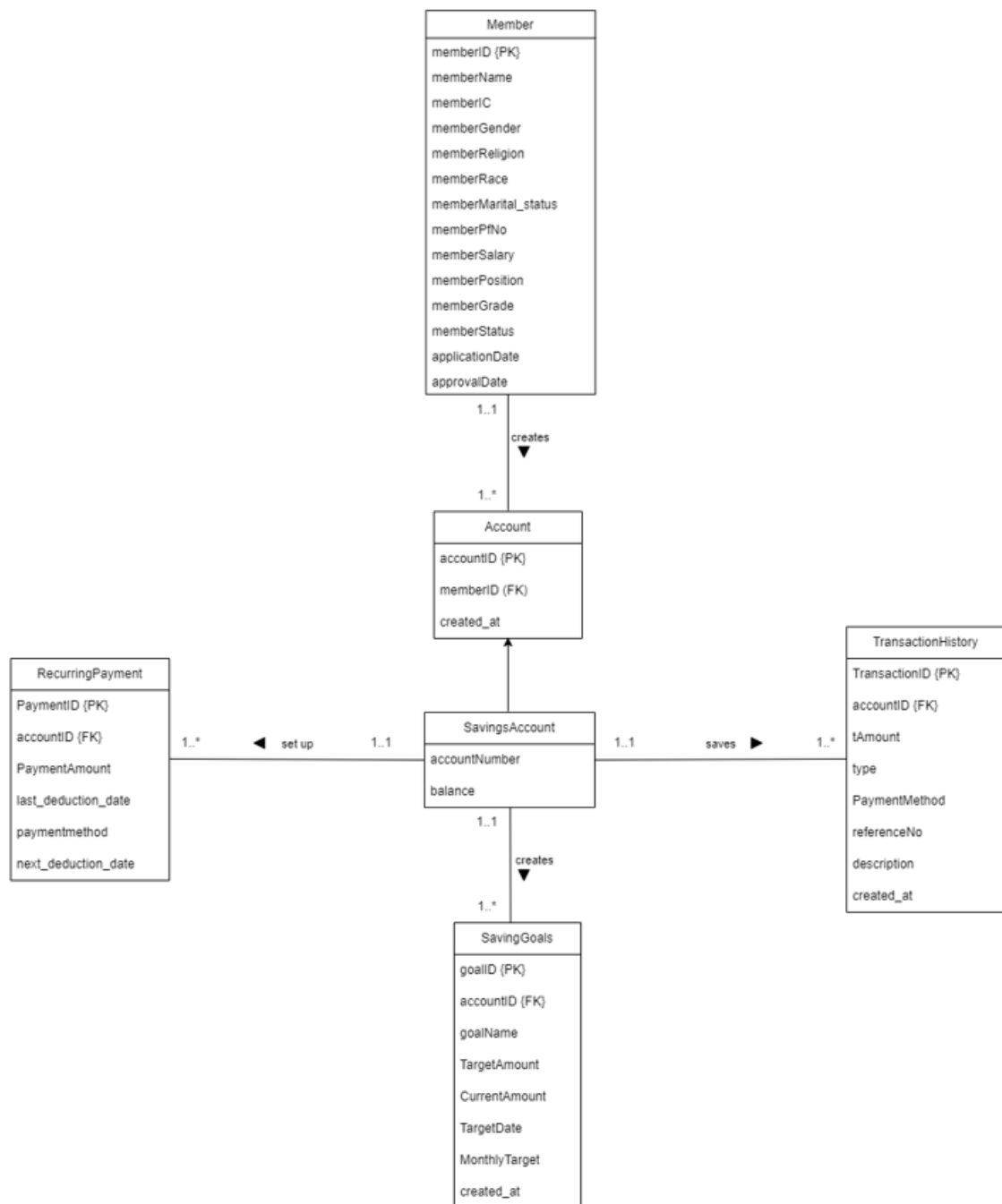
- **Automated Savings:** Members can set up recurring payments for savings.
- **Set Saving Goals:** Member can set saving goals which can update progress according to you saving amount.
- **User-Friendly Interface:** Accessible via web and mobile devices.
- **Dynamic Member Engagement:** Features like transaction history and notifications.

SECTION B (DATABASE PLANNING AND DESIGN)

2.1 Conceptual ERD



2.2 Logical ERD



2.3 Data Dictionary for Logical ERD

2.3.1 Identify Entity

Entity Name	Description	Aliases	Occurrence
Member	Represents a registered member in the system	User, Individual, Member	Multiple members linked to accounts, families, and reporting entities.
Account	Represents the financial accounts for members	User Account	Associated with saving accounts and loan accounts
Savings Account	Represents a member's savings account	Deposit Account	Linked to recurring payments, saving goals, and transaction histories
Saving Goal	Represents a member's savings goals	Financial Goal	Multiple goals linked to a savings account
Recurring Payment	Tracks recurring payment details for a member	Scheduled Payment	Multiple payments linked to savings accounts
Transaction History	Records transactions performed by members	Transactions	Multiple records per account

2.3.2 Attributes Description

Entity Name	Attributes	Data Type & Length	Description
Member	memberID (PK)	int	Member's ID
	memberName	varchar (255)	Member's name
	memberIC	varchar(22)	Member's IC
	memberGender	enum('Male', 'Female')	Member's gender
	memberReligion	varchar(50)	Member's Religion
	memberRace	varchar(50)	Member's Race
	memberMarital_status	enum('Single', 'Married', 'Divorced', 'Widowed')	Member's marital status
	memberPFNo	int	Member's PF no
	memberSalary	decimal(10,2)	Member's salary

	memberPosition	varchar(100)	Member's position
	memberGrade	varchar(50)	Member's grade
	memberStatus	enum('Pending','Lulus','Tolak')	Member's membership registration status
	applicationDate	timestamp	Application date of the membership registration
	approvalDate	timestamp	Approval date of the membership registration
Account	accountID (PK)	int	Account's ID
	memberID (FK)	int	Member's ID(refer Member entity)
	created_at	timestamp	Timestamp of when the account was created
SavingsAccount	accountNumber	Varchar(20)	Saving account's number
	balance	decimal(10,2)	Current balance in the saving account
RecurringPayment	PaymentID (PK)	int	Recurring payment's ID
	accountID (FK)	int	Account's ID (refer Account entity)
	PaymentAmount	decimal(10,2)	Amount of the payment
	last_deduction_date	date	Date of final deduction
	paymentmethod	varchar (100)	Method used for payment
	next_deduction_date	date	Date for next deduction
SavingGoals	goalID (PK)	varchar (100)	ID of the saving goal
	accountID (FK)	int	Account's ID (refer Account entity)
	goalName	varchar (100)	Name of the saving goal
	TargetAmount	decimal(10,2)	Target amount of saving
	CurrentAmount	decimal(10,2)	Current amount of saving
	TargetDate	date	Target date to achieve the saving goal
	MonthlyTarget	decimal(10,2)	Monthly target of the saving goal
	created_at	timestamp	Timestamp of when the saving goal was created
TransactionHistory	TransactionID (PK)	int	Transaction history's ID
	AccountID (FK)	int	Account's ID (refer Account entity)

	tAmount	decimal(10,2)	Transaction's amount
	type	enum("deposit", "withdrawal", "transfer_in", "transfer_out")	Type of transaction
	PaymentMethod	enum("bank_transfer", "salary_deduction", "fpx", "credit_card")	Payment method of transaction
	referenceNo	varchar (100)	Reference no of transaction
	description	text	Description of transaction
	created_at	timestamp	Timestamp of when the transaction was made

2.4 Relational Database Schema

Entity Name	Attributes
Members	memberID, memberName, memberIC, memberGender, memberReligion, memberRace, memberMarital_status, memberPFNo, memberSalary, memberPosition, memberGrade, memberStatus, applicationDate, approvalDate
Accounts	accountID, memberID, created_at
SavingsAccount	accountID, accountNumber, balance
RecurringPayments	paymentID, accountID, PaymentAmount, last_deduction_date, paymentmethod, next_deduction_date
SavingGoals	goalID, accountID, goalName, TargetAmount, CurrentAmount, TargetDate, MonthlyTarget, created_at
Transactions	TransactionID, AccountID, tAmount, type, PaymentMethod, referenceNo, description, created_at

Relation Between Entity

Entity Name	Multiplicity	Relationship	Entity Name	Multiplicity
Member	1..1	creates	Account	1..*
Savings Account	1..1	setup	RecurringPayment	1..*
	1..1	creates	SavingGoals	1..*
	1..1	saves	TransactionHistory	1..*

- A **Member** can have multiple **Account**
- A **Saving Account** can setup multiple **Recurring Payment**
- A **Saving Account** can create multiple **Saving Goals**
- A **Saving Account** can save multiple **Transaction History**

2.5 Normalization

2.5.1 Member Relation

1NF Table:

- Satisfied, have primary key
- **Primary key:** memberID

Member Table:

memberID	memberName	memberIC	memberGender	memberReligion	memberRace	memberMarital_status	memberPFNo	memberSalary

memberPosition	memberGrade	memberStatus	applicationDate	approvalDate

2NF Table: The table is in 1NF and has a single-column primary key (memberID), so all non-key attributes are fully functionally dependent on the primary key.

3NF Table: No transitive dependencies because applicationDate and approvalDate are dependent on memberID only.

Final Form:

Member Table:

memberID	memberName	memberIC	memberGender	memberReligion	memberRace	memberMarital_status	memberPFNo	memberSalary

memberPosition	memberGrade	memberStatus	applicationDate	approvalDate

Primary Key: memberID

Relation:

- **Members** (memberID, memberName, memberIC, memberGender, memberReligion, memberRace, memberMarital_status, memberPFNo, memberSalary, memberPosition, memberGrade, memberStatus, applicationDate, approvalDate)

2.5.2 Account Relation

1NF Table:

- Satisfied, have primary key
- **Primary Key:** accountID
- **Foreign key:** memberID

Account Table:

accountID	memberID	created_at

2NF Table: All non-key attributes are fully functionally dependent on accountID.

3NF Table: No transitive dependencies because depends on accountID only.

Final Form:

Account Table:

accountID	memberID	created_at

Primary Key: accountID

Foreign Key: memberID

Relation:

- **Accounts** (accountID, memberID, created_at)

2.5.3 Saving Account Relation

1NF Table:

- Satisfied, have primary key
- **Primary Key:** accountID
- **Foreign Key:** accountID

Account Table:

accountID	accountNumber	balance

2NF Table: All non-key attributes are fully functionally dependent on accountID.

3NF Table: No transitive dependencies because depends on accountID only.

Final Form:

Account Table:

accountID	accountNumber	balance

Primary Key: accountID

Foreign Key: accountID

Relation:

- **SavingsAccount** (accountID, accountNumber, balance)

2.5.4 Recurring Payments Relation

1NF Table:

- Satisfied, have primary key
- **Primary Key:** paymentID
- **Foreign Key:** accountID

Recurring Payments Table:

paymentID	accountID	PaymentAmount	last_deduction_date	paymentmethod	next_deduction_date

2NF Table:

Recurring Payments Table:

paymentID	accountID	PaymentAmount

Primary Key: paymentID

Foreign Key: accountID

Payment Duration Table:

paymentmethod	last_deduction_date	next_deduction_date

Primary Key: paymentmethod

3NF Table:

Recurring Payments Table:

paymentID	accountID	paymentmethod

Foreign Key: paymentID, accountID, paymentmethod

Payment Duration Table:

paymentID	last_deduction_date	next_deduction_date

Primary Key: paymentID

Payment Details Table:

PaymentAmount	paymentmethod

Primary Key: paymentmethod

Final Form:

Recurring Payments Table:

paymentID	accountID	paymentmethod

Foreign Key: paymentID, accountID, paymentmethod

Payment Duration Table:

paymentID	last_deduction_date	next_deduction_date

Primary Key: paymentID

Payment Details Table:

PaymentAmount	paymentmethod

Primary Key: paymentmethod

Relation:

- **RecurringPayments** (paymentID, accountID, PaymentAmount, last_deduction_date, paymentmethod, next_deduction_date)
- **PaymentDuration**(paymentID, last_deduction_date, next_deduction_date)
- **PaymentDetails**(PaymentAmount, paymentmethod)

2.5.5 Savings Goals Relation

1NF Table:

- Satisfied, have primary key
- **Primary key:** goalID
- **Foreign Key:** accountID

Savings Goals Table:

goalID	accountID	goalName	TargetAmount	CurrentAmount	TargetDate	MonthlyTarget	created_at

2NF Table: All non-key attributes are fully functionally dependent on goalID.

3NF Table: No transitive dependencies because dependent on goalID only.

Final Form:

Savings Goals Table:

goalID	accountID	goalName	TargetAmount	CurrentAmount	TargetDate	MonthlyTarget	created_at

Primary Key: goalID

Relation:

- **SavingGoals** (goalID, accountID, goalName, TargetAmount, CurrentAmount, TargetDate, MonthlyTarget, created_at)

2.5.6 Transaction Relation

1NF Table:

- Satisfied, have primary key
- **Primary key:** TransactionID
- **Foreign key:** accountID

Transaction Table:

TransactionID	accountID	tAmount	type	PaymentMethod	referenceNo	description	created_at

2NF Table:

Transaction Amount Table:

TransactionID	accountID	tAmount

Primary Key: TransactionID

Foreign Key: accountID

Transaction Reference Table:

referenceNo	type	PaymentMethod	description	created_at

Primary Key: referenceNo

3NF Table:

Transaction Table:

TransactionID	accountID	referenceNo

Foreign Key: TransactionID, accountID, referenceNo

Transaction Details Table:

TransactionID	tAmount	description	created_at

Primary Key: TransactionID

Transaction Type Table:

referenceNo	type	PaymentMethod

Primary Key: referenceNo

Final Form:

Transaction Table:

TransactionID	accountID	referenceNo

Foreign Key: TransactionID, accountID, referenceNo

Transaction Details Table:

TransactionID	tAmount	description	created_at

Primary Key: TransactionID

Transaction Type Table:

referenceNo	type	PaymentMethod

Primary Key: referenceNo

Relation:

- **Transaction** (TransactionID, accountID, referenceNo)
- **TransactionDetails** (TransactionID, tAmount, description, created_at)
- **TransactionType** (referenceNo, type, PaymentMethod)